LESSON PLAN

EJECTION SEAT (36)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to thermo nuclear weapons
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to thermo nuclear weapons. Fission – fusion and fission –fusion –fission process
5. Specific Objectives	36.1.1.1 Trainee will be able to explain conventional and nuclear explosion

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of theoretical diagrams; no hands-on practical activity due to safety restrictions
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between a conventional and a nuclear explosion?
(b) Can you define the term 'thermonuclear' in the context of weapons?
(c) Which two fundamental nuclear processes are involved in thermonuclear weapons?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Conventional vs Nuclear Explosion Mechanism • Fission Reaction Basics and Chain Reaction • Fusion Reaction Basics and Conditions	L & Ds	PPT, IB	15 min
(b)	• Fission-Fusion-Fission Staging Process • Role of Primary and Secondary Stages • Energy Release Comparison	L & Ds	PPT, PDF	20 min
(c)	• Safety Implications of Nuclear Detonation • Radiation Effects Overview • Summary of Weapon Types	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the sequence of events in a fission-fusion-fission weapon.
(b) How does the energy yield of a thermonuclear weapon compare to a conventional one?
(c) What is the specific role of the primary stage in initiating the secondary stage?

12. Home Assignment and Follow Up	Prepare a brief note on the safety arrangements required when handling nuclear weapons, to be discussed in the next lesson plan (LP 05).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety arrangements in cooperated in nuclear weapons	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Are there specific classified diagrams in the IB that cannot be displayed during the PPT session?

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min